Nithin Chakravarthi Bandi Naga

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SUMMARY

Computer Science graduate student with experience building scalable software systems, cloud-based platforms, real-time applications, and embedded systems. Seeking Summer 2026 internship opportunities focused on production-grade engineering and data-driven systems.

EDUCATION

Master of Engineering in Computer Science **Spring 2027**
University of Cincinnati, OH

Bachelor of Technology in Computer Science & Engineering (Honors, AI Specialization) **Spring 2025**
KL University, India
GPA: 9.48 / 10

TECHNICAL SKILLS AND CERTIFICATIONS

Languages: Java, Python, SQL, Bash
Backend & Real-Time: Node.js, REST APIs, WebSockets, FastAPI
AI / GenAI: TensorFlow, Retrieval-Augmented Generation (RAG), Vector Search, Agent Orchestration, OpenCV
Cloud & Systems: AWS, Docker, Kubernetes, PostgreSQL, MongoDB, CockroachDB, Neo4j, Git, Linux/Unix
Certifications: AWS Certified Solutions Architect - Associate (Jan 2024), TensorFlow Developer Certificate (Apr 2024)

PROFESSIONAL EXPERIENCE

AI/ML Intern - Mobius Networks Ltd. **Aug 2024 - July 2025**

- Designed and implemented core components of an Agent Orchestration Framework (AOF) to automate multi-step AI workflows across internal company tools.
- Built a Flowise-based orchestration layer with modular APIs to route tasks across services and models in agent-driven systems.
- Developed scalable REST APIs and real-time data pipelines using Node.js and Python to support production AI workflows.

Research Intern - IERDC, India **Dec 2021 - Aug 2024**

- Engineered embedded and IoT systems over a 2.5-year tenure, spanning telecommunications, smart campus infrastructure, and autonomous mobility initiatives.
- Developed a private telecommunication platform serving 300+ hostel students, implementing restricted-call enforcement logic, user authentication, RFID-based recharge workflows, and a web-based management interface.
- Built real-time firmware and control logic for electric and autonomous vehicle prototypes, integrating sensor modules and optimizing on-device processing.

PROJECTS

Hybrid Multi-Retriever Orchestration for Long-Term Conversational Memory **Oct 2025 - Dec 2025**

- Designed a dynamic LLM-based orchestration system selecting among five retrievers (BM25, TF-IDF, FAISS, SVM, Time-Weighted) for long-term conversational memory.
- Evaluated on LongMemEval (500 queries) and LOCOMO (1,538 queries), achieving **Recall@5 of 0.83** on LongMemEval (**18% improvement over baseline**) using Qwen2.5-7B.
- Demonstrated consistent retrieval gains across lexical, semantic, and temporal query categories, validating hybrid orchestration over fixed retrievers.

Public Addressing System (Node.js, SQL, WebSockets), *Research Project* **Nov 2023 - May 2024**

- Built a WiFi-based public addressing platform enabling real-time audio broadcasting to 200+ ESP32-controlled speaker nodes.
- Implemented a low-latency WebSocket-based system with a centralized control dashboard, enabling real-time announcements, device management, and 99.9% uptime while eliminating traditional amplifiers.